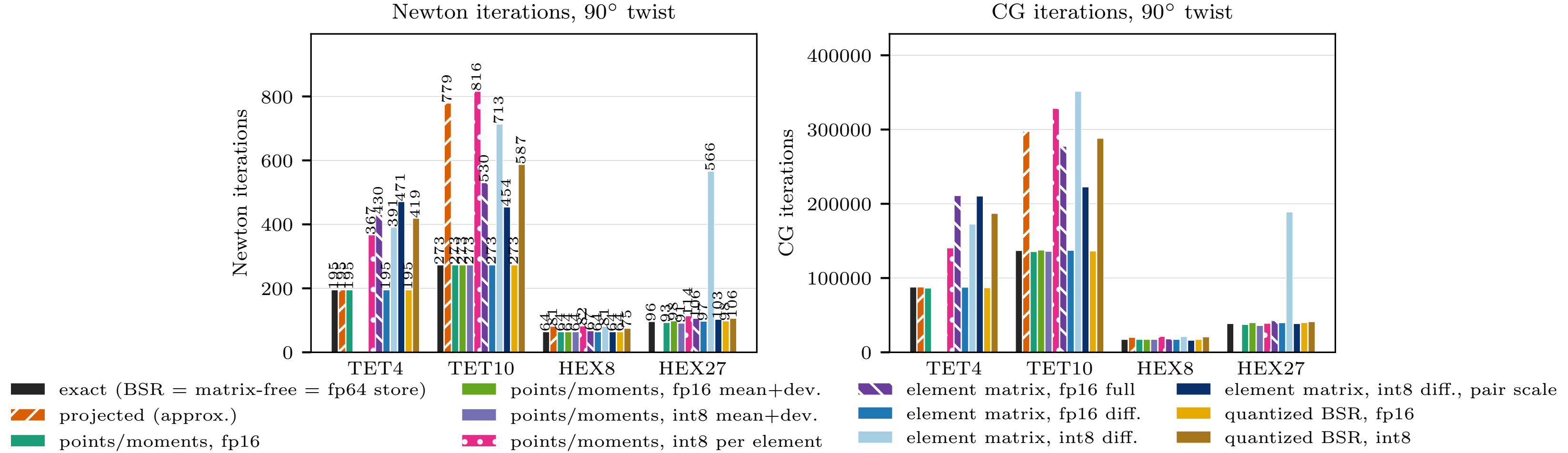
Newton iterations, 90° twist
Newton iterations
0
200
400
600
800
TET4
TET10
HEX8
HEX27
195
195
195
367
430
195
391
471
195
419
273
779
273
273
273
816
530
273
713
454
273
587
64
81
64
64
64
82
67
64
81
64
64
75
96
93
98
91
114
106
97
566
103
98
106
CG iterations, 90° twist
CG iterations
0
100000
200000
300000
400000
TET4
TET10
HEX8
HEX27
exact (BSR = matrix-free = fp64 store)
projected (approx.)
points/moments, fp16
points/moments, fp16 mean+dev.
points/moments, int8 mean+dev.
points/moments, int8 per element
element matrix, fp16 full
element matrix, fp16 diff.
element matrix, int8 diff.
element matrix, int8 diff., pair scale
quantized BSR, fp16
quantized BSR, int8